		$E'_{net} = \sqrt{E_W^2 + E_Y^2} = \sqrt{\left(\frac{kQ}{r^2}\right)^2 + \left(\frac{kQ}{r^2}\right)^2} = 1.414\frac{kQ}{r^2}$
21	**C**	$\frac{V_S}{V_P} = \frac{N_S}{N_P} \Rightarrow r.m.s.\ V_S = \frac{1600}{200} \times 25 = 200\,V$ r.m.s. current in secondary = 200 / 150 = 1.333 A mean resistor power = V^2 / R = 200^2 / 150 = 266.7 W = mean power provided by supply
22	**B**	$Q = It = 2.5 \times 10^{-6} \times 200 = 5.0 \times 10^{-4}$ C $C = Q / V = 5.0 \times 10^{-4} / 5 = 100 \times 10^{-6}$ F = 100 μF